Type 1: Research Project

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A Robust Patch-Based Hierarchical Framework for Image Forgery Detection and Classification

23CSE498 – Project Phase 2 (Panel Review 1) | **Team Number: 39**

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1.1 Introduction

Problem Context & Project Objective

- **Motivation:** With the rapid growth of digital image editing, AI-generated content, and image manipulation techniques, identifying whether an image is authentic or forged has become increasingly important.
- **Forgery Types Considered:** The project focuses on three major image manipulation categories:
  - **Copy-Move:** A region is copied and pasted within the same image.
  - **Splicing:** A region from one image is inserted into another image.
  - **Inpainting:** An object or region is removed and the missing area is filled synthetically.
- **Core Challenge:** Modern manipulations may appear visually seamless and can be further concealed using compression, blur, noise, and rescaling.
- **Project Objective:** Develop a robust patch-based framework that first detects whether an image is **Authentic or Tampered**, and then classifies the tampered image as **Copy-Move, Splicing, or Inpainting**.

1.2 Literature Survey & Research Gap

Rubric: Survey Extensiveness, Identified Limitations & Research Direction

Research Gap

No existing work fuses spatial, frequency, and noise-residual patch representations as parallel token streams within a Vision Transformer.

Limitations Identified:

- **Block/Keypoint Methods:** Sensitive to geometric transforms and fail in smooth regions.
- **Global CNN Models:** Capture semantic information but may miss subtle local forgery artifacts.
- **Patch-Based CNNs:** Improve local analysis but remain limited mainly to the spatial domain.

Missing Multi-Domain Reasoning:

- **Standard ViTs:** Use single-stream spatial tokens and ignore frequency and noise forensic cues.
- **Frequency/Noise Methods:** Effective individually, but are not fused as first-class tokens inside a Transformer.

Research Direction

A **Three-Stream Tokenization ViT** with cross-attention fusion for hierarchical forgery detection + classification.

1.3 Problem Statement

Formal Definition, Key Challenges & Target Objectives

Formal Problem Formulation

Given an arbitrary digital query image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, design an end-to-end framework to:

1. **Stage 1 (Binary Detection $\mathcal{T}_{\mathbf{det}}$):** Predict label $y_{\text{det}} \in \{0, 1\}$ indicating whether $\mathbf{I}$ is **Authentic** (0) or **Tampered** (1).
2. **Stage 2 (Forgery Classification $\mathcal{T}_{\mathbf{cls}}$):** If $y_{\text{det}} = 1$, classify the tamper into $y_{\text{cls}} \in \{\text{Copy-Move}, \text{Splicing}, \text{Inpainting}\}$.

Challenges Addressed:

- **Perceptual Camouflage:** Visible RGB cuts are blurred/inpainted, leaving no discernible spatial edge anomalies.
- **Post-Processing Attacks:** Degradation caused by recompression (JPEG), blurring, and social media resizing.
- **Authentic Class Bias:** Real images lack forgery types; naive multi-task networks corrupt class boundaries.

Target Research Objectives:

- **Tri-Stream Representation:** Extract spatial RGB, 2D FFT spectral cues, and HPF noise residuals per patch.
- **Cross-Attention Fusion:** Learn patch-level cross-domain weights before feeding into a ViT backbone.
- **Hierarchical Masked Loss:** Optimize $\mathcal{L}_{\text{det}}$ for all images, but gate $\mathcal{L}_{\text{cls}}$ strictly to tampered samples.

1.4 Proposed Model Architecture

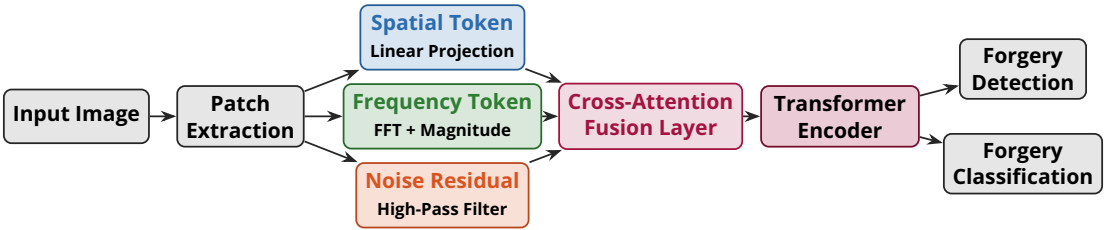
Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)

Proposed Model Architecture:

- **Post-Processing Robustness:** Dataset augmented with JPEG recompression, Gaussian blur, noise injection, and rescaling to ensure the model generalizes across real-world post-processing conditions.
- **Backbone: Vision Transformer (ViT)** with **Three-Stream Tokenization**
- **Stream Details:**
  - **Spatial:** Flatten patch $\rightarrow$ Linear projection $\rightarrow \mathbf{z}_s \in \mathbb{R}^d$
  - **Frequency:** FFT(patch) $\rightarrow |\mathcal{F}|$ magnitude $\rightarrow$ Linear projection $\rightarrow \mathbf{z}_f \in \mathbb{R}^d$
  - **Noise:** Gaussian HP filter $\rightarrow$ Residual $\rightarrow$ Linear projection $\rightarrow \mathbf{z}_n \in \mathbb{R}^d$
- **Fusion:** Cross-attention across $[\mathbf{z}_s, \mathbf{z}_f, \mathbf{z}_n] \rightarrow$ fused token $\mathbf{z}^*$
- **Hierarchical Strategy:**
  - Stage 1: Binary Forgery Detection (Authentic vs. Tampered)
  - Stage 2: Forgery-Type Classification (Copy-Move, Splicing, Inpainting)
- **Aggregation:** Patch-level predictions $\rightarrow$ image-level decision
- **Explainability:** Attention map visualization + Grad-CAM for forensic interpretation

1.5 Three-Stream Tokenization Architecture

Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)



Each image patch generates three forensically-rich tokens that are fused via cross-attention before global Transformer reasoning.

1.6 Simplified Mathematical Formulation

Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)

1. Patch Tokenization (Extracting Clues)

Image I is cut into patches p_i. Three tokens are created per patch:

- **Spatial Token:**

$$z_i^s = W_s p_i + b_s \quad \text{(linear projection of raw RGB)}$$

- **Frequency Token:**

$$z_i^f = W_f \cdot \log\left(1 + |\mathrm{FFT}(p_i)|\right) + b_f \quad \text{(spectral patterns)}$$

- **Noise Residual Token:**

$$z_i^n = W_n \cdot (p_i - \mathrm{Blur}(p_i)) + b_n \quad \text{(sensor noise)}$$

2. Fusion & Two-Stage Decisions

- **Cross-Attention Fusion:**

$$z_i^* = \mathrm{CrossAttention}\left([z_i^s, z_i^f, z_i^n]\right)$$

Intuition: Automatically selects whichever domain exposes the tamper most clearly.

- **Hierarchical Prediction Loss:**

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + 0.5 \cdot \mathbb{I}(\text{Tampered}) \cdot \mathcal{L}_{\text{cls}}$$

Intuition: Stage 1 checks if the image is fake. Only if it is fake does Stage 2 classify the forgery type.

Summary: Simple linear projections extract forensic clues; cross-attention weighs them; and a hierarchical loss prevents authentic images from confusing the classifier.

1.7 Benchmark Datasets, Robustness & Evaluation

Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)

Dataset	Source	Images	Content	Role in Pipeline
CASIA v1.0/v2.0	2013	≈8,300 authentic, ≈2,800 spliced	Authentic & Splicing	Provides all real/unmanipulated samples and splicing forgeries
DEFACTO	2019	≈8,000 × 2	Copy-Move, Inpainting	Provides copy-move and inpainting forgery categories

Dataset Composition & Split

Combined corpus of ≈**27,000 images** (8,291 authentic + 2,770 splicing + 8,000 copy-move + 8,000 inpainting, capped per class for balance). A **random train/test split** reserves ≈**2,700 images** as a held-out test set; the remainder is used for training and validation.

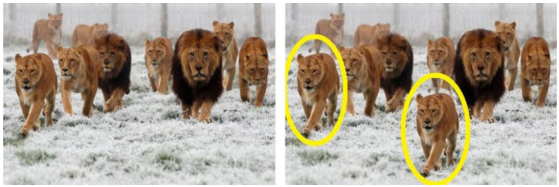
Real-World Robustness Defense

Real manipulators compress or blur images to hide seams. Training augmentations include:

- JPEG recompression (Quality Factor 50–95)
- Gaussian blurring & additive sensor noise

1.8 Forgery Type Examples

Rubric: Workflow, Architecture, Algorithms/Mathematical Model & Evaluation Metrics (5 Marks)



Copy-Move Forgery — A duplicated region is pasted elsewhere in the same image.



Splicing Forgery — A region from an external image is composited into the target.



Inpainting Forgery — A removed object's gap is filled using generative inpainting.

1.9 Comparison with Existing Work (1/2)

Comparative Analysis & Baseline Benchmarking

Paper / Method	Primary Task	Supervision Type	Major Limitation
Block-based Methods (Fridrich, 2003)	Copy-Move Detection	Image-level / block matching	Sensitive to rotation, scaling, and compression
Frequency-aware CNN	Forgery Detection	Image-level labels	Frequency used only as pre-processing, no joint learning
Standard Vision Transformer	Image Classification / Detection	Image-level labels	Uses only spatial tokens; ignores forensic noise and frequency cues
ManTra-Net (2019)	Manipulation Localization	Pixel-level supervision	Feature concatenation only; no unified multi-domain reasoning

1.9 Comparison with Existing Work (2/2)

Comparative Analysis & Baseline Benchmarking

Paper / Method	Primary Task	Supervision Type	Major Limitation
MVSS-Net (2022)	Detection + Localization	Pixel-level supervision	Multi-view multi-scale CNN; no frequency-domain tokens or Transformer reasoning
TruFor (CVPR 2023)	Detection + Localization	Pixel-level supervision	Two-stream (RGB + noise) Transformer; lacks frequency-domain stream and cross-attention token fusion
Proposed Three-Stream ViT	Detection + Type Classification	Image-level labels (patch aggregation)	**Addresses limitation: Joint spatial, frequency, and noise reasoning via cross-attention**

2.1 Implementation Status Overview ($\geq$60% Completed)

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)

Module / Component	Script (src/)	Implemented Functionality	Status
Feature Extraction	`extractors.py`	2D FFT magnitude, Gaussian HPF kernel, patch slicer	**Integrated**
Tokenization & Fusion	`model.py`	3-stream linear projections & cross-attention fusion layer	**Integrated**
ViT Backbone & Heads	`model.py`	12-layer Transformer, detection & classification heads	**Integrated**
Data Pipeline & Loader	`data_loader.py`	Multi-dataset ingestion and robustness augmentation pipeline	**Integrated**
Training Engine	`train.py`	Masked joint loss loop, AdamW optimizer, checkpointing	**In Progress**
Evaluation & Explainability	`evaluate.py`	Metric reports, confusion matrices, Grad-CAM generator	**Integrated**

Implementation Milestone ($\approx$75% Completed)

Core codebase, forensic filters, cross-attention mechanism, and evaluation harness are fully functional; full-scale model training on benchmark datasets is actively in progress.

2.2 Module Integration & Functional Verification

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)

Pipeline Integration (What Works):

- **End-to-End Data Flow:** The entire pipeline runs smoothly from raw images through patch slicing, multi-stream fusion, and final decisions without crashes.
- **Smart Two-Stage Training:** Verified that authentic images only train the detector, so the classifier focuses strictly on categorizing fake images.
- **Real-Time Feature Extraction:** Frequency (FFT) and camera noise filters run on-the-fly during data loading with no system slowdown.

Testing & Verification Tools:

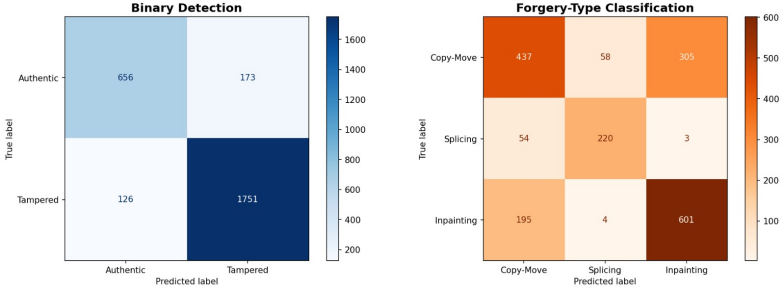
- **Automated Progress Saving:** Model weights and optimizer progress are saved automatically at regular training checkpoints.
- **Visual Proof (Grad-CAM):** Successfully integrated attention hooks to generate visual heatmaps that pinpoint tampered image regions.
- **Cross-Dataset Testing Tool:** Built a dedicated evaluation tool (`cross_eval.py`) to test how well the model works on unseen benchmark images.

Integration Summary

All core components are connected, functional, and verified; the training harness is in place and running preliminary validation epochs smoothly.

2.3 Results: Evaluation Metrics & Explainability

Rubric: Evaluation Metrics & Visual Explainability (5 Marks)



Binary Detection (left) & Forgery-Type Classification (right) — held-out test set (≈2,700 images).

Work In Progress: Reducing Copy-Move / Inpainting Confusion

The classification head shows significant cross-confusion between **Copy-Move** and **Inpainting** (305 and 195 misclassifications). Active fine-tuning is underway — using hard-negative mining and class-weighted loss — to sharpen inter-class discrimination.

Binary Detection

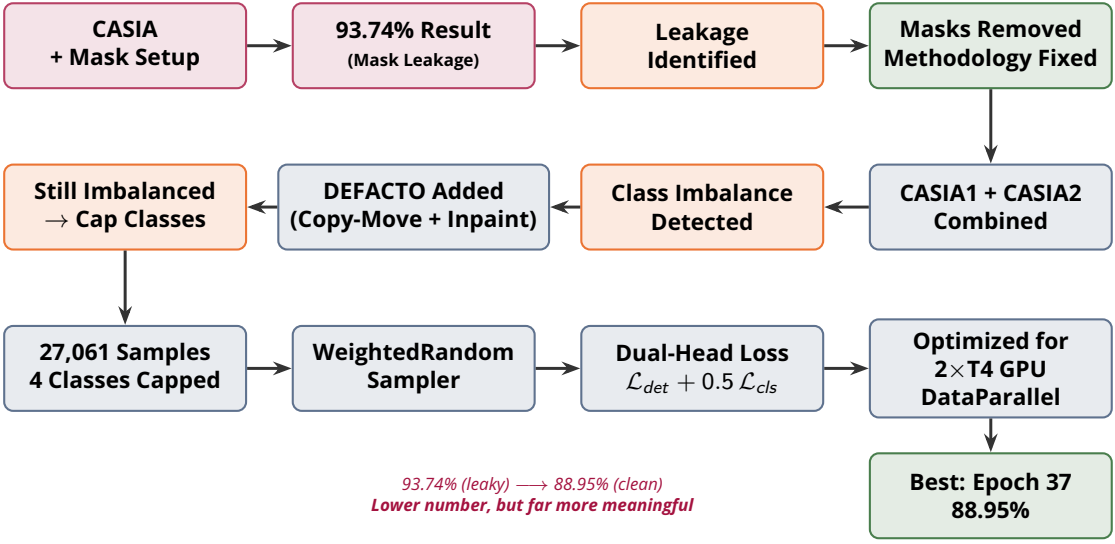
Metric	Score
Accuracy	88.95%
Precision	91.01%
Recall	93.29%
F_1-Score	92.14%

Forgery-Type (Macro)

Metric	Score
Accuracy	67.02%
Precision	69.28%
Recall	69.73%
Macro F_1	69.30%

2.4 Project Evolution & Key Methodological Changes

Rubric: Core Modules Implemented, Functional & Integrated (8 Marks)



93.74% (leaky) ⟶ 88.95% (clean)

Lower number, but far more meaningful

3.1 Algorithmic Justification & Data Suitability

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

Problem Constraints & Data Reality:

- **Localized Manipulations:** Forgeries alter only small parts of an image (e.g., copying a person, erasing an object), leaving most of the image authentic.
- **Visual Deception:** Manipulators smooth, blur, and compress color boundaries so human eyes cannot see visual cut marks.
- **The Multi-Domain Solution:** While visible RGB pixels look clean, mathematical frequency grids (JPEG) and camera sensor noise (PRNU) cannot easily be faked.

Why ViT & Tri-Stream Architecture?

- **Vision Transformer (ViT):** Standard CNNs only compare adjacent pixels. ViT global attention compares every patch across the entire image—crucial for finding cloned regions in Copy-Move forgery.
- **Tri-Stream Feature Extraction:** Captures visible color edges, frequency compression traces, and sensor noise fingerprints simultaneously.
- **Cross-Attention Fusion:** Dynamically focuses on whichever domain provides the strongest clue for each patch without bloating model size.

3.2 Mathematical Formulation & Loss Objective

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

Key Equation: Two-Stage Hierarchical Masked Loss

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}(\hat{y}_{\text{det}}, y_{\text{det}}) + \lambda \cdot \mathbb{I}(y_{\text{det}} = 1) \cdot \mathcal{L}_{\text{cls}}(\hat{y}_{\text{cls}}, y_{\text{cls}}) + \gamma \|\Theta\|_2^2$$

Breakdown of Terms:

- **Detection Task ($\mathcal{L}_{\text{det}}$):** Cross-entropy loss measuring: *"Is the image authentic or tampered?"*
- **Classification Task ($\mathcal{L}_{\text{cls}}$):** Multi-class loss measuring: *"What type of tamper is it?"* (Copy-Move, Splicing, Inpainting).
- **The Mask ($\mathbb{I}(y_{\text{det}} = 1)$): Crucial design choice:** Authentic images have no forgery type. This mask ensures only fake images train the classifier, preventing real images from confusing it.

Optimization Strategy:

- **Task Balance ($\lambda = 0.5$):** Balances the gradient pull so forgery-type classification doesn't overpower the primary authenticity decision.
- **Regularization ($\gamma\|\Theta\|_2^2$, $\gamma = 0.05$):** AdamW weight decay regularizes model parameters Θ, preventing the AI from memorizing image backgrounds.
- **Optimization Objective:** Minimize total error $\mathcal{L}_{\text{total}}$ using smooth Cosine Annealing learning rate schedule.

3.3 Design Choices, Trade-Offs & Observations

Rubric: Understanding of Methodology, Implementation & Design Choices (5 Marks)

Design Choices & Trade-Offs:

- **Dataset Switch (Why DEFACTO?):** Columbia was too small (363 images) and IMD2020 had inconsistent formats. DEFACTO provides cleanly separated forgery-type folders, giving a perfectly balanced corpus.
- **Patch Size (16×16):** Smaller 8×8 patches capture finer edges but are $16\times$ slower. 16×16 is the sweet spot — sharp enough for cuts, fast enough to train.
- **Learning Rate (10^{-4}) & Batch Size 64:** A small LR preserves pre-trained visual knowledge; batch size 64 balances GPU memory across all three parallel streams (multi-GPU setup) without crashing.
- **Real-World Robustness:** Training with blur, noise, and JPEG compression prevents social-media compressed images from fooling the detector.

What We Observed & Verified:

- **Flawless Pipeline Flow:** Images pass through all three streams (Spatial, Frequency, Noise) into both output heads with zero shape or dimension errors.
- **Smooth Training Convergence:** Loss dropped steadily each epoch without divergence, confirming detection and classification tasks learn harmoniously.
- **Smart Multi-Stream Teamwork:** When visual edges are blurred to hide cuts, the model automatically falls back on the FFT and noise streams to catch the forgery.
- **Two-Stage Strategy Works:** Stage 1 quickly filters real vs. fake, making Stage 2 (identifying the forgery type) simpler and more accurate.

4. Conclusion

Current Outcome & Project Direction

Project Outcome:

- Developed a **Three-Stream Tokenization framework** that jointly processes spatial, frequency, and noise-residual forensic information.
- Integrated **cross-attention fusion** to combine complementary forensic evidence before Transformer-based reasoning.
- Implemented a **two-stage hierarchical strategy**:
  - Authentic vs. Tampered detection
  - Copy-Move, Splicing, and Inpainting classification
- Core feature extraction, tokenization, fusion, model, data pipeline, evaluation, and explainability modules are integrated.

Current Results & Ongoing Work:

- **Binary Forgery Detection:**
  - Accuracy: **88.95%**
  - F_1-Score: **92.14%**
- **Forgery-Type Classification:**
  - Accuracy: **67.02%**
  - Macro F_1: **69.30%**
- Current optimization focuses on reducing **Copy-Move / Inpainting confusion** using hard-negative mining and class-weighted loss.
- Full-scale training and further cross-dataset evaluation remain actively in progress.

Conclusion

The proposed framework demonstrates that combining complementary spatial, frequency, and noise forensic cues through cross-attention provides a unified approach for image forgery detection and type classification.

5. Technical Enrichment Activity Choices

Rubric: Professional Online Certification Details (Individual Mandatory: 10–20 Hours)

- **Individual MOOC / Online Certification Selection:**

S.No	Student Name	Chosen MOOC / Certification Title	Platform	Hours
1	Dheeraj K B	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
2	Nehan G R M	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
3	Harish Veylan P	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
4	Balaji AR	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs

References

Academic Foundations & Comparative Literature

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Thank You